

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 15

Algorithmic Complexity and the Search for Generative Equations

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. What is the fundamental difference between finding that a dataset is well-fit by $y = ax^2 + bx + c$ and finding that the system generating the data is governed by $F = ma$?

2. You compress a file using gzip and then try to compress it again. The second compression achieves almost nothing. Why? State this in terms of Kolmogorov complexity.

3. A random string is maximally complex in the Kolmogorov sense. Is this the opposite of what you would expect? Explain why randomness = maximum complexity.

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4. You are searching for the equation governing a physical system. You try $\{+, -, \times, \div, \sin, \cos, \exp\}$ as your primitive operations and find nothing good. What would it mean to “expand the search space”? What are the risks?

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5. A vendor demos a static-analysis tool and claims it “detects every infinite loop in any codebase, with no false alarms.” Which result from this chapter says the claim, as stated, is impossible? Practical tools that catch many infinite loops do exist, name two relaxations of the vendor’s claim that make it achievable, and say what each relaxation trades away.

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
